Telepathic UX: The Next Generation of Mobile Apps

Microsoft

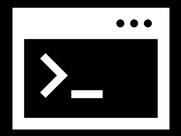


David Ortinau

Principal Product Manager

Microsoft

3 UI Paradigms



You tell the computer exactly how



Point and click to direct how

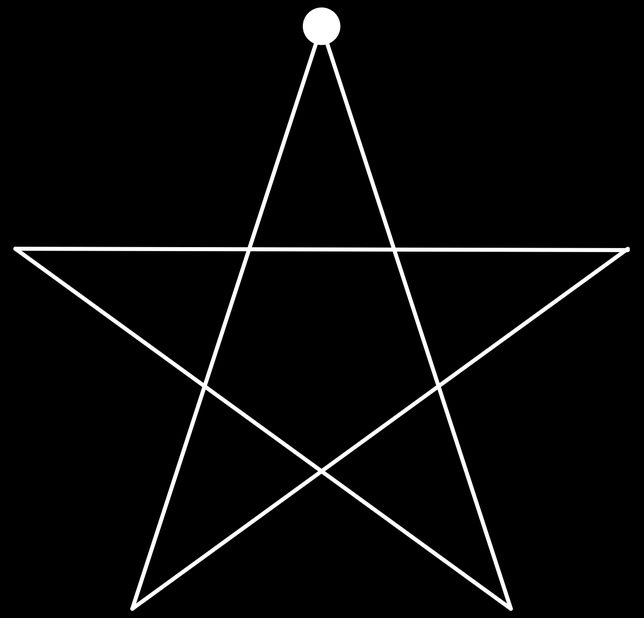


Tell the computer what you want

01 · 1967 · LOGO

Drawing by telling.

The turtle moves where the code points. A drawing is a program.

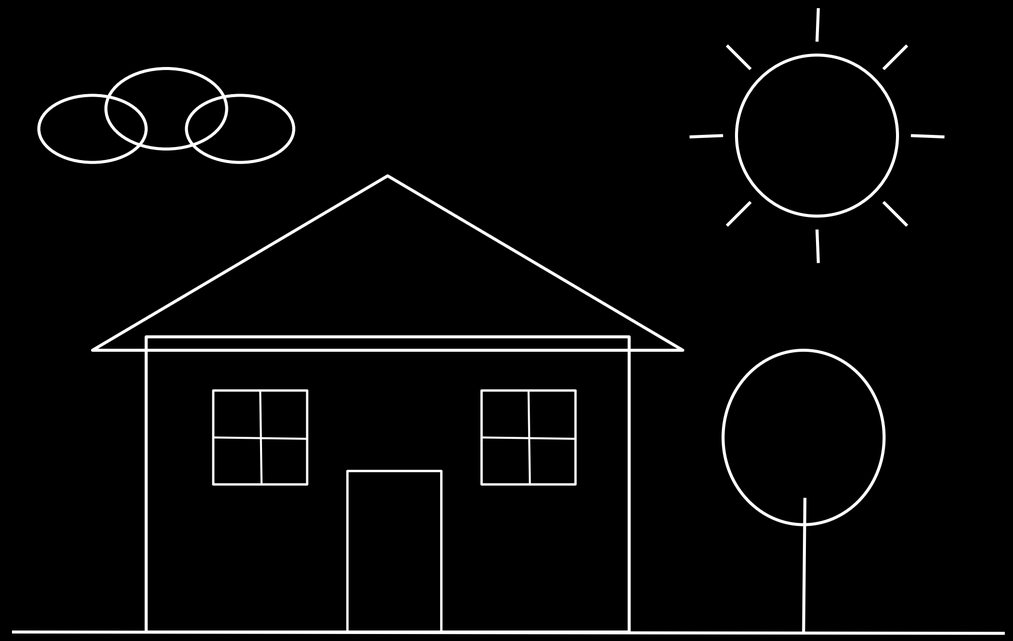


```
PENDOWN FORWARD 100 RIGHT 144 FORWARD 100 RIGHT 144
FORWARD 100 RIGHT 144 FORWARD 100 RIGHT 144 FORWARD 100
RIGHT 144 PENUP
```

02 · 1984 · MACPAINT

Drawing by hand.

The cursor becomes a brush. Pixels respond to motion — the screen becomes a canvas.

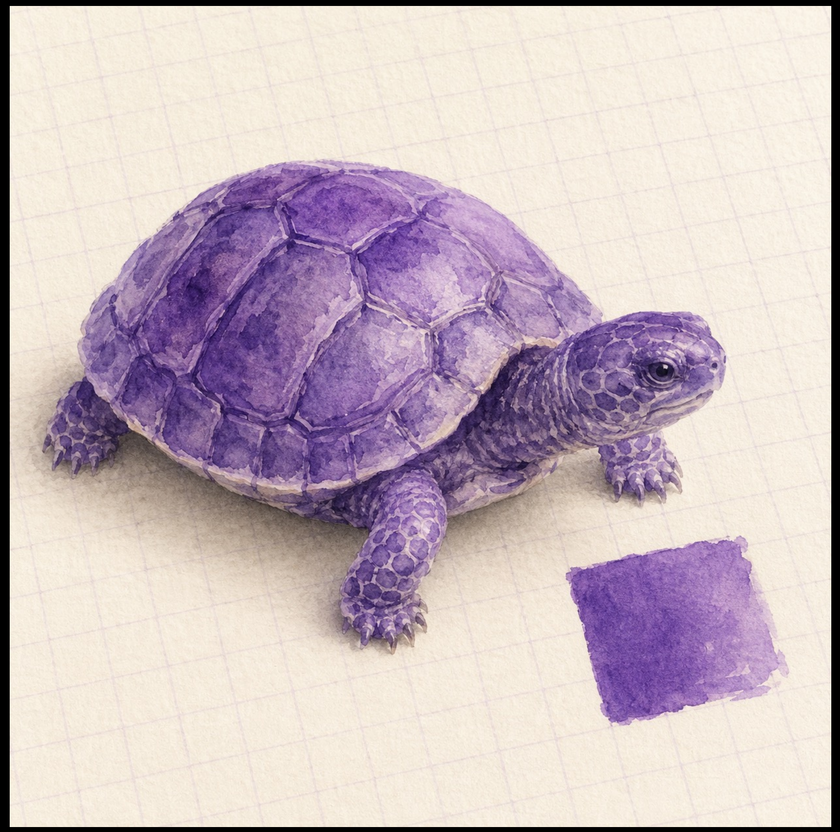


select ◆ brush ◆ bucket ◆ pencil

03 · 2024 · AI

Drawing by asking.

You describe the intent. The model proposes the image — and you steer the next one.



“a watercolor of a turtle drawing a square on graph paper”

“AI is introducing the third user-interface paradigm in computing history, shifting to a new interaction mechanism where users tell the computer what they want, not how to do it — thus reversing the locus of control.”

Jakob Nielsen

00 Demo

Building blocks

What powers the Barista Notes experience

INPUT 01



MediaElement

Camera and photo capture for multimodal input

INPUT 02



Community Toolkit

Speech-to-text for hands-free voice input

AI 03



Microsoft.Extensions.AI

Unified AI abstractions across providers and models

AI 04



Microsoft.Maui.Essentials.AI

On-device Apple Intelligence on iOS and macOS

AI 05



Azure.AI.OpenAI

Cloud LLM for reasoning and content generation

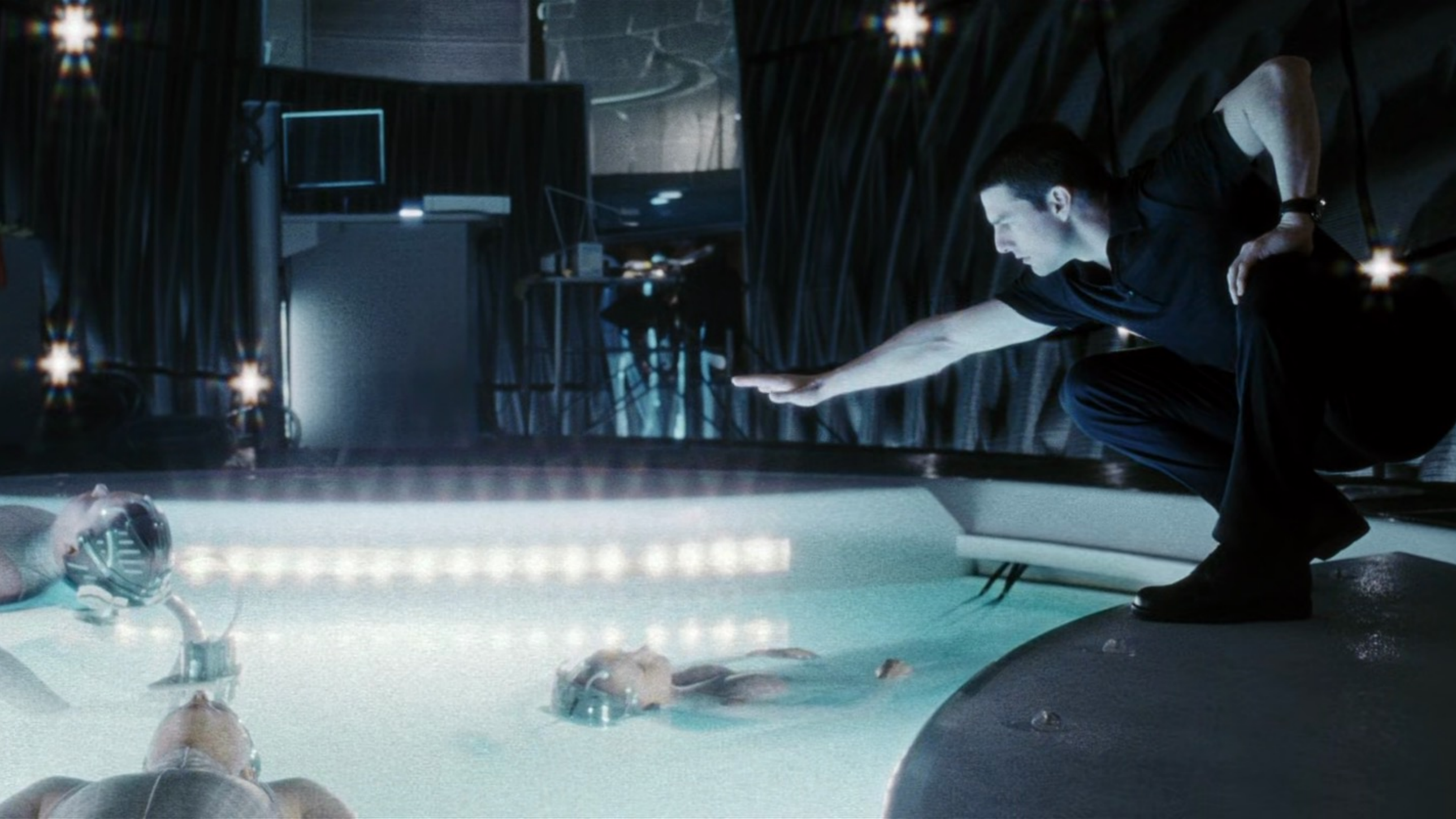
INTEGRATION 06



Microsoft.Maui.AI.Attributes

Tool calling and AI-driven navigation

01 A Vision of Precognition



precognition
From Latin

prae-
(prefix) before, in front of,
prior to.

e.g. praehistorio -> pre-history

cognitio, cognitionis (f.)
(noun) knowledge,
learning, recognition

e.g. cognition Veritatis -> knowledge of the truth

"Generative AI emerges as a key driver in reshaping user interfaces, introducing new possibilities for personalized, multimodal, and cross-platform interactions.

8ieniek et al., 2024
Generative AI in Multimodal User Interfaces: Trends, Challenges, and Cross-Platform Adaptability

AI-generated UIs and proactive adjustments:

Generative AI offers a transformative potential to create UIs that are dynamically tailored to real-time user behavior and contextual cues, leading to more personalized and efficient interaction spaces. By analyzing user habits, preferences, and situational factors, generative AI models can adapt UIs to prioritize relevant information, simplify navigation, and reduce cognitive load. This enables systems to provide user-specific experiences that evolve based on interaction patterns, ultimately enhancing accessibility and usability.

Predictive modeling is another key innovation driving proactive UI adjustments. Leveraging past user interactions, these models can anticipate user needs, making adjustments to interface components such as shortcuts, layout, and displayed content before the user initiates a task. This proactive approach streamlines workflows, minimizes navigation time, and provides a more intuitive interaction experience, especially valuable in professional environments like design, healthcare, and engineering, where efficiency is critical.

Emotionally adaptive UIs and autonomous collaboration: Emotionally adaptive UIs represent a frontier in responsive design, enabling interfaces to adjust based on real-time assessments of user emotional states. By analyzing multimodal inputs-such as facial expressions, tone of voice, and physiological cues-these UIs can modify design elements (e.g., color scheme, layout) and functionality to create a more supportive and comfortable experience. This capability is particularly beneficial in applications like mental health support and customer service, where responding to user emotions can significantly improve engagement and satisfaction.

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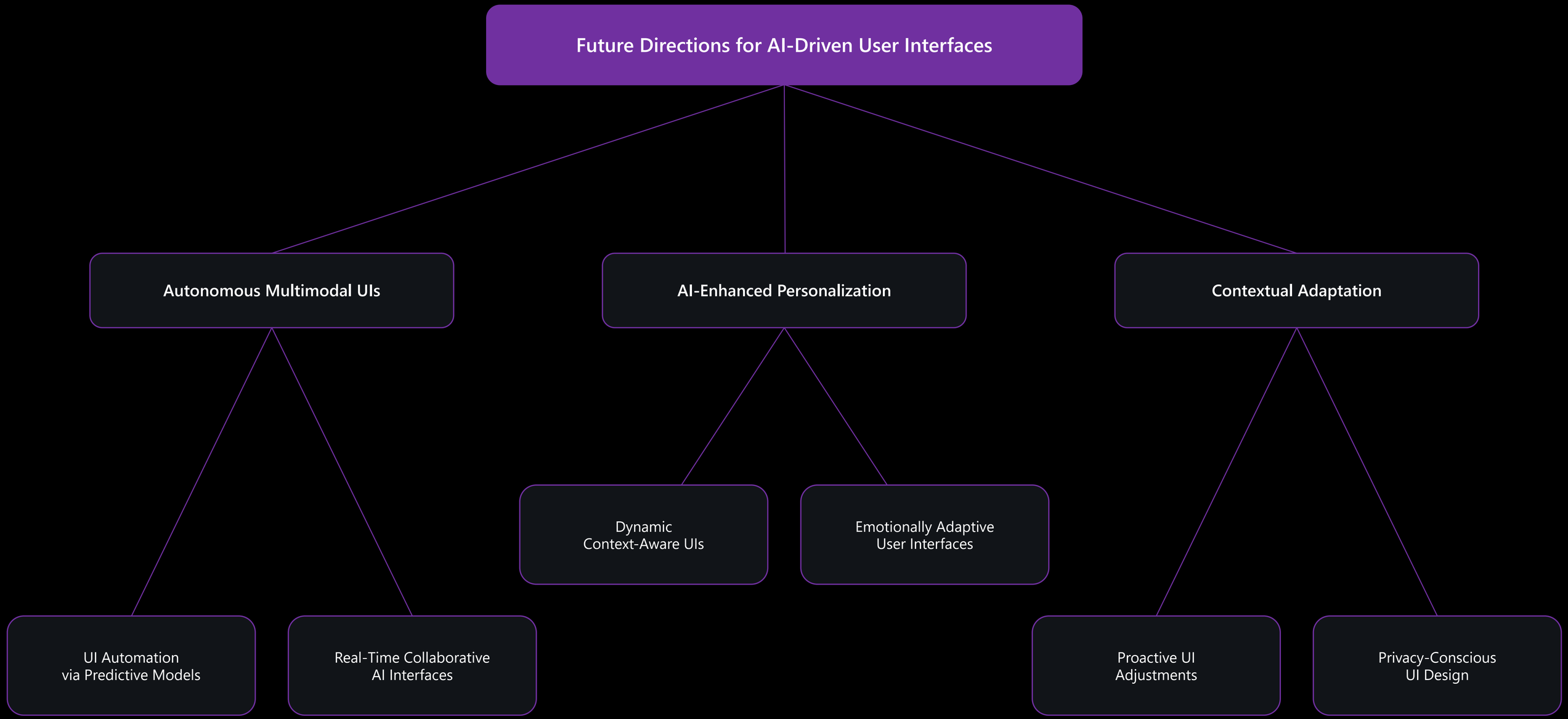
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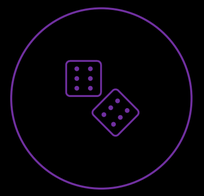
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Future Directions for AI-Driven User Interfaces
Autonomous Multimodal UIs
AI-Enhanced Personalization
Contextual Adaptation
UI Automation via Predictive Models
Real-Time Collaborative AI Interfaces
Dynamic Context-Aware UIs
Emotionally Adaptive User Interfaces
Proactive UI Adjustments
Privacy-Conscious UI Design

Six Principles for Generative AI Design

Weisz, He, Muller, Hoefer, Miles & Geyer · CHI 2024

NEW CHARACTERISTICS



01 Generative Variability

Same input, different outputs. Help users navigate, compare, and curate across multiple results.



02 Co-Creation

Provide controls — prompts, parameters, co-editing — so users can shape what the model produces.



03 Imperfection

Outputs may be wrong or unwanted. Make uncertainty visible, offer ways to fix or regenerate.

NEW INTERPRETATIONS



04 Responsibly

Human-centered approach, surfaced value tensions, governed emergent behaviors, tested for harms.



05 Mental Models

Teach users how the system behaves; learn how they think about it. Build the model in both directions.



06 Appropriate Trust & Reliance

Calibrate trust with explanations and rationales. Use friction to slow users at key decisions.

02 What We Need

Context

Building Context
.NET MAUI



Location

GPS and proximity are very powerful triggers for pieces of information.



Calendar

When granted access this resource can provide a ton of context relevant to decision making for the user.



Contacts

Essentials provides quick access to the device contacts store.



Memory

User behavior can be tracked and recorded for AI to identify useful patterns that can be predictive of future behaviors. Kept securely on-device, this information can be used to create a behavioral snapshot.



Sensors

Sensors tell you if the user is moving, moving fast, in a quiet or noisy environment, if it's light or dark around them.



Platform Specifics

Some platforms expose additional information like ambient light and biometrics.

03 Demos

Thank you



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